

PART 2 MOTION

- (b) which component changes uniformly with time?
 (c) If the initial horizontal velocity of the projectile is decreased, will the time taken for the projectile to fall to the ground be decreased, increased, or remain the same?

2 When an aeroplane is at an altitude of 490 m and moving horizontally at a speed of 160 m s^{-1} , a small object is dropped from it. The following questions refer to the period from immediately after the instant the object is dropped till just before it hits the ground.

- (a) What is the vertical component of the object's initial velocity?
 (b) What is the vertical component of its displacement?
 (c) What is the vertical component of its acceleration?
 (d) Deduce the time it is in flight.
 (e) What is the horizontal component of its initial velocity?
 (f) What is the horizontal component of its final velocity?
 (g) What is the horizontal component of its acceleration?
 (h) Deduce the horizontal component of its displacement.

3 A ball is thrown horizontally with a speed of 14.0 m s^{-1} from a point 6.40 m above the ground. Calculate

- (a) the time taken by the ball to reach ground level;
 (b) the horizontal distance travelled in that time;
 (c) its velocity when it reaches the ground.

4 A stone is projected out to sea from the edge of a vertical cliff 122.5 m high, the velocity of projection being 7.00 m s^{-1} horizontally. Find

- (a) the time the stone takes to reach the water;
 (b) the distance from the base of the cliff at which it strikes the water;
 (c) the velocity at that instant.

5 Two stones are thrown simultaneously in line and horizontally from the top of a cliff 78.4 m high with speeds 5.00 m s^{-1} and 20.0 m s^{-1} respectively. How far apart will they strike the water? Find the velocity of the first stone as it reaches the water.

6 A bomber travelling horizontally with a speed of 576 km h^{-1} at an altitude of 10 000 m drops a bomb.

- (a) Taking the origin of a Cartesian coordinate system as the point at which the bomb was dropped and the x -direction as that in which the plane was travelling, deduce two equations which give the position of the bomb at time t after it was released. (x and y will represent distances in metres, and t the time in seconds. The two equations will give x and y as functions of t .)
 (b) Deduce the equation which describes the path of the bomb while in flight.
 (c) What is the name of the shape of the path followed by the bomb?

7 A tennis player hits a lob shot at point X in Figure 13.1 with a velocity of 13 m s^{-1} at an angle of 60° to the horizontal.

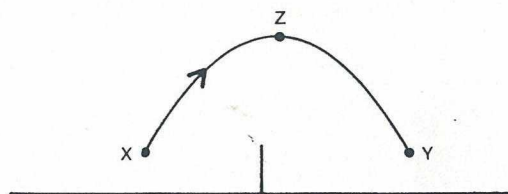


Fig. 13.1

Answer the following questions for the flight of the ball from X to a point Y, the same height above the ground as X.

- (a) What is the vertical component of the ball's initial velocity?
 (b) What is the vertical component of the ball's velocity at Z, the highest point of its flight?
 (c) What is the vertical component of the ball's acceleration as it moves from X to Z?
 (d) How long does the ball take to travel from X to Z?
 (e) How long does the ball take to travel from X to Y?
 (f) What is the horizontal component of the ball's velocity at X?
 (g) What is the horizontal component of the ball's velocity at Z?
 (h) What is the horizontal distance from X to Y?

8 A stone is thrown with a velocity of 20 m s^{-1} at an angle of 40° with the horizontal. Determine

- (a) the time for which it is above the level at which it is thrown;
 (b) the greatest vertical height reached;
 (c) its range on a horizontal plane.

9 At the XXI Olympiad in Montreal in the Women's High Jump a new Olympic record was set by Rosemarie Ackermann of East Germany.

- (a) If 0.44 s elapsed between the instant she left the ground and the time she passed over the bar, what vertical height did she raise her centre of mass?
 (b) If her centre of mass was 1.10 m above the ground at take-off and 0.12 m above the bar as she cleared it, what was the new Olympic record?

10 A missile is projected with a velocity of 30.0 m s^{-1} at an angle of 60.0° with the horizontal. At what height will it strike a cliff 45.0 m distant?

11 A cricketer hits a six, the ball having travelled 100 m horizontally when it strikes the ground beyond the boundary. The greatest height reached by the ball is 25 m. Calculate the velocity with which the ball left the bat.

12 A boy standing on the tray of a utility travelling at 18 km h^{-1} throws a ball with a velocity of 7.35 m s^{-1} vertically upwards (relative to himself) and catches it again at the same level. What horizontal distance does the ball move while it is in the air?

ANSWERS

- 12 (a) 1.00×10^{-2} N downwards (b) 2.45×10^{-2} N
(c) 1.45×10^{-2} N upwards
13 (a) L (b) L (c) L (e) S
14 Shorter time

SET 13 PROJECTILE MOTION NEAR THE EARTH'S SURFACE

- 1 (a) Horizontal component (b) Vertical component
(c) Remains the same
2 (a) Zero (b) 490 m downwards (c) 9.80 m s^{-2} downwards (d) 10.0 s (e) 160 m s^{-1} in the direction in which the plane was travelling (f) 160 m s^{-1} in the direction in which the plane was travelling (g) Zero (h) 1.60 km
3 (a) 1.14 s (b) 16.0 m (c) 17.9 m s^{-1} at an angle of 38.7° with the horizontal
4 (a) 5.00 s (b) 35.0 m (c) 49.5 m s^{-1} at an angle of 81.9° to the horizontal
5 60.0 m; 39.5 m s^{-1} at an angle of 82.7° to the horizontal
6 (a) $x = 160t$; $y = -4.90 t^2$ (b) $y = -1.91 \times 10^{-4} x^2$ (c) A parabola
7 (a) 11 m s^{-1} (b) Zero (c) 9.8 m s^{-2} downwards (d) 1.1 s (e) 2.2 s (f) 6.5 m s^{-1} (g) 6.5 m s^{-1} (h) 14 or 15 m
8 (a) 2.6 s (b) 8.4 m (c) 40 m
9 (a) 0.95 m (b) 1.93 m
10 33.8 m
11 31 m s^{-1} at an angle of 45° to the horizontal
12 7.5 m
14 (Note that the stone is thrown at an angle of 45° to the horizontal to achieve the maximum horizontal range) Greatest vertical height is $\frac{1}{2}d$
15 (a) S (b) S (c) L (d) H (e) S (f) L (g) L (it will be negative)
16 (a) S (b) S (c) L (d) L (f) S (g) L (h) L
17 A longer horizontal distance during the rising part of the flight

SET 14 UPHILL AND DOWNHILL MOTION

- 1 (a) 245 N (b) 245 N (c) 424 N
2 1.13 N
3 1.1×10^3 N
4 (a) Zero (b) 0.22 N
5 (a) 96 N downhill (b) 82 N (c) 18 N
6 1.17×10^3 N
7 0.32° or $0^\circ 19'$
8 1.4 m s^{-2}

SET 15 MOTION IN A CIRCLE

- 1 (a) 8.0 m s^{-2} (b) 8.0×10^3 N (c) Towards the centre of the circular path
2 2.0×10^4 N
3 0.034 m s^{-2}
4 (a) 40 Hz (b) $1.2 \times 10^4 \text{ m s}^{-2}$
5 4.5 Hz
6 1.26×10^3 m
7 (a) 6.0 m s^{-1} along the tangent at A (b) 6.0 m s^{-1} along the tangent at B (c) 0.35 s (d) 6.0 m s^{-1} at an angle of 60° with the tangent at B (e) 17 m s^{-2} (f) 29 N
8 6.3 s
9 (a) 3.0 m s^{-2} north (b) 4.0 m s^{-2} west (c) 5.0 m s^{-2} west 37° north (d) 8.0×10^2 N west 37° north
10 (b) 1.5 m (c) 0.92 N towards the centre of the circular path (d) 1.0 N (e) 1.8 s
11 (a) 1.5 (b) 1.5 (c) 6.0
12 41°
13 8°
14 (a) (i) 15.0 N downwards (ii) 15.0 N upwards (b) (i) 10.1 N downwards (ii) 19.9 N upwards
15 (a) 70.0 m s^{-1} (b) 2.0×10^3 N upwards
16 (a) 329 N upwards (b) 1.20×10^4 N upwards

SET 16 SIMPLE HARMONIC MOTION

- 1 (a) 25 N upwards (b) 9.0 m s^{-2} downwards
2 (a) 6.0 N m^{-1} (b) 1.8 N to the left (c) 0.40 m to the left (d) $1.0 \times 10^2 \text{ m s}^{-2}$ to the right
3 0.8 s
4 (a) 5.0 N m^{-1} (b) 1.0 N to the right (c) Zero (d) 0.50 N to the left
5 (a) 114 N (b) 57.0 N
6 (a) C (b) A and E (c) 1.0 m s^{-1} (d) 0.87 m s^{-1}
7 1.61 m s^{-1}
8 0.693 m s^{-1}
9 0.359 s ; 1.05 m s^{-1}
11 $2\pi\sqrt{(L-h)/g}$
12 0.90 s
13 $5h \text{ m}$; $4\pi\sqrt{h/g}$
14 1.27 s
15 1.678×10^{-2} N
16 44.937 s per day gain
17 (a) $v = \omega R \cos \omega t$ (b) $a = -\omega^2 R \sin \omega t$ (c) $F = -m\omega^2 R \sin \omega t$
18 (a) 0.10 m (b) 0.5 Hz (c) (i) $v = 0.10\pi \cos \pi t$ (ii) $a = -0.10\pi^2 \sin \pi t$ (d) 0.31 (or $\pi/10$) m s^{-1} (e) 0.58 m s^{-2}